COURSE SYLLABUS

CMPT 470/816: Advanced Software Engineering

Time and Location: Tues/Thurs 10:00–11:20 am, Arts Building 134



Department of Computer Science, University of Saskatchewan

176 Thorvaldson Building, 110 Science Place, Saskatoon, SK, S7N 5C9, Canada

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1 Course Information

1.1 Catalog Description

Covers advanced software engineering principles and techniques. Includes software architecture, software evolution, reverse engineering, design recovery, refactoring, software comprehension, software analysis, domain-specific techniques, requirements and specifications, advanced design and modeling techniques, formal methods, and the business of software.

1.2 Course Prerequisite(s) and Logistics

Prerequisite(s): CMPT 214.3 and CMPT 370.3.
Note: Students with credit for CMPT 816 will not receive credit for this course.

Time and Location: Tues/Thurs 10:00–11:20 am, Arts Building 134 (in-person)
Course website: Canvas
Tutorials: There are no tutorials for this course. Students are however encouraged to approach the instructor for any problems.

1.3 Course Instructor(s) and TAs

Prof. Chanchal Roy, PhD
Office: THORV 281.4
Office hours: By email/appointment
Email: chanchal.roy@usask.ca
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1.4 Learning Outcomes

By the end of this course, students are expected to:

1. have a comprehensive understanding of advanced software engineering and how AI is revolutionizing this field. They will learn foundational principles, cutting-edge AI tools, and emerging trends shaping the future of software systems.
2. learn TXL, a powerful tree-based programming language for transforming and analyzing code, and tackle real-world software engineering challenges through examples and tasks.
3. acquire a deep understanding of traditional and automated software development processes, release engineering, and DevOps, including their challenges and opportunities, to better navigate and optimize the software lifecycle.
4. learn mining software repositories, such as GitHub, BigQuery, Stack Overflow, and OpenAI developers forums, through code extraction, processing, and validating large-scale software data. They will also develop skills to analyze software data and apply software metrics effectively.
5. build expertise in change management, including change impact analysis and the application of AI-driven solutions, to address software evolution and maintenance effectively.
6. adopt industry-leading practices of modern software engineering, such as integration of AI and automation, low-code/no-code development, and Cloud-Native Development to optimize and future-proof software engineering workflows.
7. understand and apply AI and Software Engineering (AI4SE, SE4AI) to automate tasks such as code generation, code clone detection, software migration, software testing, and program repair.
8. learn prompt engineering techniques, principles of trustworthy and responsible AI, and methods for verifying foundation models to ensure their transparency and explainability.
9. develop proficiency in understanding and applying design patterns, including creational, structural, and behavioral patterns, to solve common software design challenges and enhance software system quality.

1.5 Course Overview

This course explores the intersection of traditional and advanced software engineering practices with modern AI-driven solutions. It provides students with both a theoretical foundation and practical hands-on experience to tackle contemporary challenges in software development, quality assurance, and maintenance. The major topics covered include:

1. **Introduction to Advanced Software Engineering.** This section sets the stage by revisiting foundational software engineering concepts and analyzing the evolving role of AI in the discipline. Students will learn about the principles that drive innovation in software development to address advanced software engineering challenges.

2. **Source Transformation with TXL.** Source transformation is a critical component of modern software engineering. This topic introduces the syntax and semantics of **TXL**,

empowering students to perform tasks such as code normalization, optimization, and language migration (e.g., converting C-based systems to Java).

❸ **Software Development Processes and Challenges.** Students will explore traditional development methodologies, including Agile processes and Extreme Programming. Emphasis is placed on process automation, such as Requirements Engineering, Software Design Automation, and Documentation Automation, to enhance productivity and minimize human errors.

❹ **Software Repository Mining and Data-Driven Engineering.** This hands-on module introduces students to Software Repository Mining tools & techniques to mine data from GitHub, BigQuery, and OpenAI Developers Forum, and so on, enabling the extraction of actionable insights from real-world data. Students will also learn to validate and build datasets to support evidence-based decision-making in software engineering.

❺ **Software Change Management.** This section will emphasize the effective management of software evolution through structured processes like Change Request Handling, Change Impact Analysis (CIA), and Version Control. Students will explore tools and methodologies for assessing and implementing changes while maintaining system integrity. The integration of AI-driven tools is highlighted, showcasing how automation can optimize code, predict risks, and streamline change processes. Practical topics such as Low-Code/No-Code Development (LCNC) and DevOps Practices, including CI/CD pipelines, are also covered, equipping students with skills to manage software changes in dynamic environments efficiently.

❻ **AI and Software Engineering (AI4SE, SE4AI).** Artificial Intelligence (AI) has emerged as a transformative force in software engineering, driving innovation across all stages of the software lifecycle. This module introduces the role of AI in addressing complex software engineering challenges, focusing on Generative AI and Large Language Models (LLMs) such as ChatGPT and Gemini. Students will explore the application of AI and LLMs in tasks such as code generation, bug fixing, large-scale software clone detection, and automated documentation. Emphasis will be placed on understanding the technical details of these models and their implications for improving software productivity, scalability, and quality. Practical insights will highlight how AI can enhance decision-making, reduce development costs, and address scalability challenges in modern software engineering workflows.

❼ **Software Quality Assurance.** Software quality assurance is a critical aspect of building reliable and efficient software systems. The course will cover various methods for ensuring quality, with a focus on modern automated testing tools and frameworks. Additionally, students will explore the verification and explainability of foundation models, along with the principles of trustworthy AI, highlighting their role in maintaining transparency, reliability, and ethical standards in software systems.

❽ **Design Patterns in Software Engineering.** Design patterns are essential tools for building reliable, scalable, and maintainable software systems. This course segment introduces key design patterns, emphasizing their role in addressing recurring design challenges and improving software architecture. Students will explore creational, structural, and behavioral patterns and understand their significance in software design.

2 Course Topics and Details

Week	Dates	Topics	Milestones
Week #01	Jan 07, 2025 Jan 09, 2025	**Introduction to Advanced Software Engineering** • Course overview and objectives **Source Transformation with TXL** • Introduction to TXL	M0: Group Formation (0 mark but required), M1: Playing with TXL (individually)
Week #02	Jan 14, 2025 Jan 16, 2025	**Software Analytics** • Mining Software Repositories • Analysis of software data, Software metrics	M2: Dataset Construction (in group)
Week #03	Jan 21, 2025 Jan 23, 2025	**AI and Software Engineering** • Introduction to AI4SE and SE4AI • Prompt Engineering Techniques for SE	
Week #04	Jan 28, 2025 Jan 30, 2025	**AI and Software Engineering** • Automating SE tasks with foundation models (code generation, code clone detection, software migration) • AI model optimization	M3: Project Proposal and Outline (in group)
Week #05	Feb 04, 2025 Feb 06, 2025	**AI and Software Engineering** • Trustworthy AI for SE • Responsible AI for SE	
Week #06	Feb 11, 2025 Feb 13, 2025	**Modern Software Engineering Practices** • Integration of AI and automation, Low code/No code (LCNC) • Cloud-native development (Docker)	M4: Initial Project Progress Report (in group)
		Mid-Term Break	
Week #07	Feb 25, 2025 Feb 27, 2025	**Software Change Management** • Initiation, concept and feature location, impact analysis • Actualization, refactoring, verification	
Week #08	Mar 04, 2025 Mar 06, 2025	**Software Development Process** • Introduction to Software Development Process • Agile processes, extreme programming, and pair programming • Release engineering and DevOps	M5: Intermediate Project Progress Report (in group)
Week #09	Mar 11, 2025 Mar 13, 2025	**Software Process Automation** • Software requirements, software design • Code Summarization, code documentation	
Week #10	Mar 18, 2025 Mar 20, 2025	**Software Quality Assurance** • Introduction to software testing, Test-Driven Development • Automated test generation, Automated Program Repair	
Week #11	Mar 25, 2025 Mar 27, 2025	**Software Quality Assurance / Project Presentation** • Verification of Foundation Models • Transparency and explainability of foundation models	M6: Final Presentation (in group)
Week #12	Apr 01, 2025 Apr 03, 2025	**Design Patterns in Software Engineering / Project Presentation** • *Introduction to Design Patterns:* Overview of design patterns and their importance in solving common software design problems. • *Categories of Design Patterns:* Exploring creational, structural, and behavioral patterns.	M6: Final Presentation (in group), M7: Final Report Submission (in group)

3 Recommended Readings/Textbooks

[1] J.R. Cordy, "**Excerpts from the TXL Cookbook**", Generative and Transformational Techniques in Software Engineering, Lecture Notes in Computer Science 6491, January 2011, pp. 27-91.

[2] Vaclav Rajlich"**Software Engineering: The Contemporary Practice**", CRC Press, Nov. 2011, 315 pages.

[3] Mark Richards, "**Software Architecture Patterns**", O'Reilly media, Inc. 2022.

[4] Martin, Robert C "**Clean code: a handbook of agile software craftsmanship**", Pearson Education, 2009.

[5] Fowler, Martin. "**Refactoring: improving the design of existing code**", Addison-Wesley Professional, 2018

[6] Eric Freeman and Elisabeth Robson, "**Head First Design Patterns**", Second Edition, O'Reilly media, Inc. 2020.

Note: Selected readings will be made available on the course Canvas website.

4 Assessment and Grading

The methods for student evaluation and the grading scheme for assessing performance in this course are detailed below.

4.1 Evaluation Components

The course workload consists of the following components:

- **Exercises and Quizzes (Required):** Conducted throughout the term to encourage active class participation.
- **Assignment (Required):** One assignment will be provided on TXL, with due dates in mid-January.
- **Exploratory Project (Required):** Students are required to complete a project.
- **Final Exam (Required for undergraduate students):** An assessment at the end of the term.

4.2 CMPT 470: Grading Scheme and Milestones

4.2.1 Grading Scheme

Table 1: CMPT 470 Grading

Deliverable	Due Date	Weighting
Exercises and Quizzes	Throughout the term	05%
M0: Group Formation	Mid January	0%
M1: Playing with TXL	Mid January	07%
M2: Dataset Construction	Late January	13%
M3: Project Proposal and Outline	Early February	05%
M4: Initial Project Progress Report	Late February	10%
M5: Intermediate Project Progress Report	Mid March	15%
M6: Final Presentation	Late March / Early April	05%
M7: Final Report Submission	Early April	20%
Final Exam	See schedule (in-person). Usually, the exam will be for *three hours*.	20%

4.2.2 Milestones

Exercises and Quizzes: There will be 5 to 6 exercises or quizzes throughout the term. These will typically be conducted in class without prior notice and will be based on the material covered during the current or most recent lecture.

M0 (Group Formation): You have to make a group of three students. If you do not, instructors or TAs will randomly assign you to a group.

M1 (Playing with TXL): Perform software analysis using source transformation systems.

Course Project: The project milestones are mandatory for undergraduate students.

- *Exploratory Project and Task:* Students will work on a group project (a group of three students is recommended) under the mentorship and guidance of instructors and TAs. They can select their own project topics, but these must be approved by the instructors. Projects can focus on areas such as software repository mining, empirical software engineering, software engineering tools and techniques, or large language models.
- *Milestones of the Exploratory Project are below.*
  - **M2 (Dataset Construction):** Perform data extraction, process the collected data and finalize the dataset for the Project. Instructors and TAs will guide and assist you in constructing your dataset.
  - **M3 (Project Proposal and Outline):** Submit a project proposal outlining your chosen topic (a template will be provided). Resources will be provided to help you conceptualize and refine your idea. All project ideas must be approved by the instructors.
  - **M4 (Initial Project Progress Report):** Submit an initial report on the project's development and progress.
  - **M5 (Intermediate Project Progress Report):** Submit an extended report on the project's development and progress.
  - **M6 (Final Presentation):** You are required to give a presentation consisting of eight minutes for presenting your project and two minutes for a Q&A session.
  - **M7 (Final Report Submission):** Deliver a comprehensive report summarizing the project outcomes and analysis (a template will be provided).
  - **Encouraged:** Regular discussions with the instructor and TAs to refine and enhance the project.
  - **Essential:** Ensure proper citation and quotation to maintain academic integrity.

Final Exam: It will be an in-person exam
Time duration: Three hours
Date, time, and location: See the schedule once announced by the Registrar.
Question Types: There will be a mix of question types, including multiple-choice, short-answer, long-answer, analytical questions, and true/false questions.

Other activities: Include demonstrating tools used in SW development and maintenance, exercises, and some unannounced quizzes.

4.3 CMPT 816: Grading Scheme and Milestones

4.3.1 Grading Scheme

Graduate students are exempt from the final exam but are required to complete an exploratory project. This project can be undertaken individually (recommended), in collaboration with other graduate students, or as part of a mixed group with undergraduate students.

Table 2: CMPT 816 Grading

Deliverable	Due Date	Weighting
Exercises and Quizzes	Throughout the term	05%
Critical Review and Replication Study	Throughout the term	08%
M1.816: Playing with TXL	Mid January	07%
M2.816: Project proposal	Third week of January	10%
M3.816: Project Outline	Third week of Feb	10%
M4.816: Project progress report	Third week of March	20%
M5.816: Final Presentation	TBD	05%
M6.816: Project Report	Third week of April	35%

4.4 Milestones

Exercises and Quizzes: There will be 5 to 6 exercises or quizzes throughout the term. These will typically be conducted in class without prior notice and will be based on the material covered during the current or most recent lecture.

M1.816 (Playing with TXL): Perform software analysis using source transformation systems.

Other activities: Include demonstrating tools used in SW development and maintenance, exercises, and some unannounced quizzes.

Course Project: The project is mandatory for graduate students. Graduate students are required to consult with the instructor regarding the project.

- *Exploratory Project and Task:* Students will investigate a software engineering topic of their choice, **subject to approval from the instructor.** The project may focus on languages, tools, processes, or formalisms. A critical synthesis and thorough analysis of the chosen topic is expected as part of the project deliverables.
- *Milestones of the Exploratory Project are below.*
  - **M2.816 (Introductory Proposal):** Submit a brief proposal (a template will be provided) outlining the project topic.
  - **M3.816 (Project Outline):** Provide an update that includes an outline of sources, preliminary findings, and a description of the project's focus.
  - **M4.816 (Project Progress Report):** Submit a report on the project's development and progress.

- **M5.816 (Final Presentation):** You are required to give a presentation consisting of ten minutes for presenting your project and two minutes for a Q&A session.
- **M6.816 (Project Report):** Deliver a comprehensive report summarizing the project outcomes and analysis (a template will be provided).
- **Encouraged:** Regular discussions with the instructor to refine and enhance the project.
- **Essential:** Ensure proper citation and quotation to maintain academic integrity.

5 Course Policies

Submitting Assignment and Project Deliverables: Assignments and project deliverables must be submitted via the course page on Canvas. Specific instructions will be provided for each assignment.

Late Submission: Students who anticipate or experience delays in submitting the assignment or project deliverables should contact the instructor. Depending on the circumstances, a mutually agreeable arrangement may be made without penalties.

Missed Submission: If an assignment or project deliverables are missed, students are encouraged to discuss the situation with the instructor. Alternative arrangements may be made when feasible.

Undergraduate Course Completion Policy Undergraduate students must attend the final exam, but passing it is not a prerequisite to pass the course. Overall grades are based on the assignment, project, quizzes, and final exam.

Attendance Expectations: Attendance is not mandatory; however, up to 8% (including up to 3% of bonus marks) of the overall grade is allocated for in-class discussions and quizzes.

Recording of the Course: Lectures will not be recorded; recording is prohibited except where AES accommodation allows.

Statement on the advent of AI/GPT: In this course, AI/GPT tools may only be used for specific course materials and project deliverables. Their use for any other parts of the course is strictly prohibited.

Integrity Defined (from the Office of the University Secretary): The University of Saskatchewan is committed to the highest standards of academic integrity and honesty. Students are expected to be familiar with these standards regarding academic honesty and to uphold the policies of the University in this respect. Students are particularly urged to familiarize themselves with the provisions of the Student Conduct & Appeals section of the University Secretary Website and avoid any behavior that could potentially result in suspicions of cheating, plagiarism, misrepresentation of facts, and/or participation in an offense. Academic dishonesty is a serious offense and can result in suspension or expulsion from the University.

All students should read and be familiar with the Regulations on Academic Student Misconduct (https://governance.usask.ca/student-conduct-appeals/academic-misconduct.php - StudentAcademicMisconductRegulations) as well as the Standard of Student Conduct in NonAcademic Matters and Procedures for Resolution of Complaints and Appeals (https://governance.usask.ca/student-conduct-appeals/non-academic-misconduct.php)

For more information on what academic integrity means for students, see the Academic Integrity section of the University Library Website at: https://library.usask.ca/academic-integrity.php

You are encouraged to complete the Academic Integrity Tutorial to understand the fundamental values of academic integrity and how to be a responsible scholar and member of the USask community - https://libguides.usask.ca/AcademicIntegrityTutorial

There are also valuable resources on the Integrity Matters: https://academicintegrity.usask.ca

Access and Equity Services (AES) for Students: Students who have disabilities (learning, medical, physical, or mental health) are strongly encouraged to register with Access and Equity Services (AES) if they have not already done so. Students who suspect they may have disabilities should contact AES for advice and referrals at any time. Those students who are registered with AES with mental health disabilities and who anticipate that they may have responses to certain course materials or topics, should discuss course content with their instructors prior to course add / drop dates. In order to access AES programs and supports, students must follow AES policy and procedures. In order to access AES programs and supports, students must follow AES policy and procedures. For more information or advice, visit https://students.usask.ca/health/centres/access-equity-services.php, or contact AES at 306-966-7273 or aes@usask.ca.

Students registered with AES may request alternative arrangements for mid-term and final examinations. Students must arrange such accommodations through AES by the stated deadlines.

Instructors shall provide the examinations for students who are being accommodated by the deadlines established by AES.

Student Supports

- *Student Learning Services.* Student Learning Services (SLS) offers assistance to U of S undergrad and graduate students. For information on specific services, please see the SLS website https://library.usask.ca/studentlearning.
- *Teaching, Learning, and Student Experience.* The Teaching, Learning, and Student Experience Unit (TLSE) focuses on providing developmental and support services and programs to students and the university community. For more information: https://students.usask.ca.
- *College Supports.* Students in Arts & Science are encouraged to contact the Undergraduate Student Office and/or the Trish Monture Centre for Success with any questions on how to choose a major; understand program requirements; choose courses; develop strategies to improve grades; understand university policies and procedures; overcome personal barriers; initiate pre-career inquiries; and identify career planning resources. Contact information is available at: (http://artsandscience.usask.ca/undergraduate/advising)

6 Treaty Acknowledgment

As we gather here today, we acknowledge we are on Treaty Six Territory and the Homeland of the Métis. We pay our respect to the First Nation and Métis ancestors of this place and reaffirm our relationship with one another.